2023 - quiz12

Graded

Student

William Chen

Total Points

10 / 10 pts

Question 1

A stationary... **5** / 5 pts

1.1 **Is the...** **2** / 2 pts

- ✔ **– 0 pts** Correct
- **– 1 pt** approach error
- **– 2 pts** wrong approach

1.2 **In terms...** **3** / 3 pts

- ✔ **– 0 pts** Correct
- **– 1 pt** approach error
- **– 2 pts** wrong approach

Question 2

The figure... **5** / 5 pts

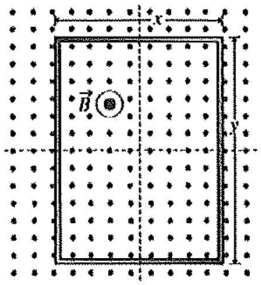
- ✔ **– 0 pts** Correct

Quiz 12

PH-UY 2023, Spring 2026
A Sections

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(1) A stationary rectangular loop (dimensions x-by-y) of total resistance R sits in a region with a uniform magnetic field of magnitude B directed out of the page. The magnetic field strength now begins to change at a steady rate, during which time a steady *counterclockwise* current I appears in the loop.



(a) (2 pts) Is the magnetic field increasing or decreasing in strength? Justify your answer.

decreasing, the current is counter clockwise so by the RHR, the induced magnetic field is directed out of the page.

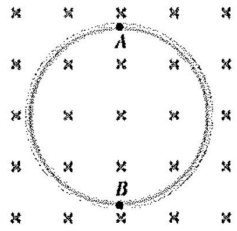
(b) (3 pts) In terms of x, y, R, and I, at what rate is the magnitude of $\vec{B}$ changing?

$\mathcal{E} = \frac{d\Phi_B}{dt}$ $\quad \frac{d\Phi_B}{dt} = xy\frac{dB}{dt}$

$I = \frac{\mathcal{E}}{R} \Rightarrow \mathcal{E} = IR$

$IR = xy\left|\frac{dB}{dt}\right|$ $\quad \left[\frac{dB}{dt} = -\frac{IR}{xy}\right]$

(2) (5 pts) The figure shows a circular loop of metal wire of radius r in a uniform magnetic field of magnitude B_0 directed into the page. You grab points A and B and pull them in opposite directions, stretching the loop vertically (and thus compressing it horizontally) until its area is very nearly zero, taking a time T to do so. What was the *average magnitude* of the induced emf in the loop during this time? Express your answer solely in terms of r, B_0, T and constants of nature, as needed.



$\mathcal{E}_{avg} = \frac{|\Delta\Phi_B|}{\Delta T}$ $\quad \Phi_B = BA$

$|\Delta\Phi_B| = B(\pi r^2)$

↳ since $A_f = 0$

$\left[\mathcal{E}_{avg} = \frac{B\pi r^2}{T}\right]$